

TOSONTSENGEL, Mongolia

WMO: 442250

Lat: 48.7631N

Lon: 98.2661E

Elev: 1723

StdP: 82.27

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.1	-41.7	-46.2	0.0	-43.6	-44.2	0.1	-41.7	7.9	-17.2	6.1	-19.6	0.1	0	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	27.2	14.7	25.2	13.8	23.4	13.1	16.2	24.3	15.1	22.7	14.1	21.2	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	11.8	18.7	12.1	10.9	17.9	11.0	10.1	17.0	52.0	24.6	48.4	22.8	45.3	21.3	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.4	5.4	DB	-45.4	30.6	3.2	2.1	-47.7	32.1	-49.6	33.3	-51.4	34.5	-53.8	36.0	
			WB	-45.5	17.6	2.9	1.8	-47.5	18.8	-49.2	19.9	-50.9	20.8	-53.0	22.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.7	-31.0	-26.5	-14.1	1.1	7.4	13.6	16.1	13.6	6.9	-2.3	-16.1	-27.0	(6)
	DBStd	17.47	6.00	6.57	7.62	5.26	4.47	3.54	2.92	3.17	4.04	5.71	7.54	6.59	(7)
	HDD10.0	5844	1272	1021	747	268	102	10	1	7	108	381	782	1147	(8)
	HDD18.3	8441	1530	1254	1006	517	338	146	82	151	343	639	1032	1405	(9)
	CDD10.0	463	0	0	0	1	23	118	188	118	15	0	0	0	(10)
	CDD18.3	17	0	0	0	0	0	3	12	2	0	0	0	0	(11)
	CDH23.3	415	0	0	0	0	15	105	215	78	2	0	0	0	(12)
	CDH26.7	77	0	0	0	0	1	14	51	11	0	0	0	0	(13)
(14) Wind	WSAvg	1.4	0.5	0.8	1.5	2.4	2.5	1.8	1.5	1.3	1.6	1.5	1.0	0.7	(14)
(15) Precipitation	PrecAvg	211	3	3	3	9	14	38	59	45	18	11	7	5	(15)
	PrecMax	315	22	12	9	29	37	85	145	94	57	47	20	21	(16)
	PrecMin	147	0	0	0	1	1	7	18	12	3	1	0	0	(17)
	PrecStd	42	4	3	2	7	9	19	25	20	12	8	5	5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.0	-4.8	8.0	18.7	25.2	28.6	30.6	28.2	23.0	15.0	5.1	-4.8	(19)
		MCWB	-11.1	-6.7	2.2	8.1	11.8	14.8	17.3	15.1	12.0	6.3	1.0	-6.5	(20)
	2%	DB	-15.1	-9.7	3.8	15.2	22.3	26.3	27.9	25.5	20.4	11.9	0.9	-9.9	(21)
		MCWB	-15.8	-10.9	-0.3	6.1	11.0	13.9	15.6	13.9	10.9	4.8	-2.3	-10.9	(22)
	5%	DB	-19.0	-13.1	0.9	12.7	19.9	24.2	25.6	23.4	18.0	9.4	-2.1	-13.5	(23)
		MCWB	-19.5	-13.9	-2.0	5.0	9.4	13.0	14.5	13.3	9.4	3.3	-4.4	-14.2	(24)
	10%	DB	-22.2	-16.2	-1.9	10.3	17.4	22.2	23.7	21.4	15.7	6.9	-5.1	-17.0	(25)
		MCWB	-22.6	-16.9	-4.1	3.8	7.9	12.1	13.7	12.6	8.3	1.8	-6.8	-17.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.3	-6.5	2.5	9.0	13.8	16.2	18.8	16.6	12.9	6.9	1.5	-6.3	(27)
		MCDB	-10.2	-4.8	7.7	17.2	23.8	25.7	28.3	25.0	21.1	14.0	5.4	-4.7	(28)
	2%	WB	-15.8	-11.0	-0.1	6.7	11.5	15.0	17.0	15.2	11.4	5.2	-1.8	-10.9	(29)
		MCDB	-15.1	-9.9	3.4	13.9	20.8	24.3	24.9	22.6	18.8	11.3	1.2	-9.9	(30)
	5%	WB	-19.5	-13.9	-2.0	5.4	9.8	13.9	15.7	14.1	10.1	3.9	-3.9	-14.1	(31)
		MCDB	-19.0	-13.1	0.8	12.0	18.8	22.1	23.1	21.0	16.9	9.4	-1.5	-13.5	(32)
	10%	WB	-22.5	-16.9	-4.1	4.0	8.4	12.9	14.6	13.2	8.7	2.3	-6.1	-17.5	(33)
		MCDB	-22.2	-16.2	-1.9	10.0	16.5	20.4	21.5	20.0	15.2	7.1	-4.3	-17.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	13.5	15.4	12.7	13.9	13.6	12.6	13.2	14.7	14.1	11.7	9.7	(35)
		MCDBR	13.0	15.0	15.2	16.4	18.8	17.8	16.8	17.4	18.8	18.2	14.6	13.1	(36)
		MCWBR	12.3	13.8	11.7	9.7	10.4	8.8	7.6	8.3	10.7	11.6	11.3	12.3	(37)
	5% WB	MCDBR	13.1	15.0	14.9	15.6	17.5	15.3	14.3	14.6	17.2	17.6	14.1	13.0	(38)
		MCWBR	12.4	13.8	11.7	9.7	10.4	8.2	7.4	7.4	10.0	11.6	11.3	12.2	(39)
(40) Clear-Sky Solar Irradiance	taub		0.221	0.245	0.286	0.325	0.330	0.334	0.337	0.319	0.286	0.267	0.245	0.226	(40)
	taud		2.343	2.295	2.265	2.321	2.345	2.432	2.459	2.479	2.521	2.494	2.407	2.335	(41)
	Ebn at Noon		879	925	931	924	929	925	917	922	928	894	837	819	(42)
	Edh at Noon		65	89	110	116	119	110	106	99	85	73	62	58	(43)
(44) All-Sky Solar Radiation	RadAvg		1.74	2.82	4.38	5.69	6.44	6.45	6.09	5.36	4.50	2.99	1.78	1.36	(44)
	RadStd		0.09	0.13	0.18	0.23	0.39	0.30	0.34	0.22	0.23	0.17	0.07	0.06	(45)

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page